

Complex real-world reinforcement learning environments present continuous state spaces that are too rich and detailed for effective high-level RL planning. My thesis project will aim to bridge the gap between low-level control and high-level reasoning by compressing these rich inputs into discrete, semantically meaningful tokens. I aim to develop a Transformer-based state encoder (leveraging concepts from VQ-VAE or similar discretization techniques), or some type of diffusion model to map continuous sensor data (e.g., visual input) into a compact, discrete vocabulary of state tokens. These tokens are intended to represent abstract, human-like concepts (e.g., "Door is open," "Object is grasped"). This work will focus on quantifying the efficiency gains realized when a planning policy operates exclusively on this symbolic, tokenized state space compared to the original, high-dimensional space, and developing visualization tools to confirm the semantic fidelity of the learned discrete representations.